Project Python Foundations:FoodHub Data Analysis

Post Graduate Program in Artificial Intelligence and Machine Learning: Business ApplicationsPGP-AIML-BA-UTA-Mar26-C

Nitesh Kumar 15th April 2026

Contents / Agenda

Executive Summary

- **Conclusion:**

The analysis of Foodhub data analysis provided key information with respect to the cuisines that are popular among users, and which restaurants are bringing in the most revenue. The analysis also signals that the users order more over the weekend than during the weekend.

- **Insight :**
    - American, Japanese, Italian, Chinese and Mexican are the most popular cuisine.
    - Cost of the order is right skewed, with 75% of orders below $22.50
    - Shake Shack is the most popular restaurant, it accounts for 11.43% of orders
    - Weekend have 2.5 times more volume compared to weekdays.

Executive Summary continued....

- **Insight :**
  - Total cost of the food order was $31314.82 with 71.51% revenue from weekend and weekdays it was only 28.49%
  - 736 out of 1898 orders don’t have ratings and which accounts for 38.78 % of the total.
  - Shake Shack is the most popular restaurant, it accounts for 11.43% of orders
  - There is very low correlation between cost of the order and delivery time or food preparation time and need to probe more and collect data to find impact of these field on revenue and customer experience
  - Average Delivery time on weekend is 22 min vs on weekdays it is 28 days.

Executive Summary continued....

- **Recommendations :**
  - **Cuisine** : Since American was the most popular cuisine, the company could target more revenue by partnering with more American restaurants
  - **Weekdays vs Weekend business**: Weekend account for most of the business, while weekdays are slow hours. Company should focus on promotions to boost weekday sales further
  - **Delivery times** : Delivery times during the weekday are more and there is a negative correlation to revenue. So, company should look at how to reduce delivery times during weekdays by partnering with more restaurants to provide more closer options, especially for the popular cuisines
  - **Ratings:** There are a lot of missing ratings, focused initiative to actively collect feedback and bridge this gap is needed.
  - **Additional Data**: Company should collect additional data with order dates/month to further analyze trends in order volume for seasonal changes, areas where most orders are coming from etc. to more accurately target partnerships with additional restaurants to increase revenue

Business Problem Overview

The number of restaurants in New York is increasing day by day. Lots of students and busy professionals rely on those restaurants due to their hectic lifestyles. Online food delivery service is a great option for them. It provides them with good food from their favorite restaurants. A food aggregator company FoodHub offers access to multiple restaurants through a single smartphone app.

The app allows restaurants to receive a direct online order from a customer. The app assigns a delivery person from the company to pick up the order after it is confirmed by the restaurant. The delivery person then uses the map to reach the restaurant and waits for the food package. Once the food package is handed over to the delivery person, he/she confirms the pick-up in the app and travels to the customer's location to deliver the food. The delivery person confirms the drop-off in the app after delivering the food package to the customer. The customer can rate the order in the app. **The food aggregator earns money by collecting a fixed margin on the delivery order from the restaurants.**

Objective

The food aggregator company has stored the data of the different orders made by the registered customers in their online portal. **The objective is to analyze the data to get a fair idea about the demand of different restaurants which will help them in enhancing their customer experience. In addition, the they want data to answer** how the company can improve its business.

Solution Approach

- **Problem Statement** - As a Data Scientist in this company, do an exploratory data analysis for the questions provided by the Data Science team to find answers to these questions that will help the company improve its business.

- **Approach -** To analyze the data and answer the questions, I will do the following steps:
  - Load the data from the file into a dataframe
  - Understand the shape, datatypes and value distributions (missing values, unique values, mean, median mode) of the dataset
  - Prep the data for analysis for e.g. fill missing values, drop any rows/outliers that may skew the analysis etc.
  - Perform Univariate Analysis on the columns, plotting them using various visual graphs
  - Perform Bivariate analysis to understand the relationships of different variables with respect to each other
  - Draw conclusions based on the analysis between restaurants with higest rating, the cuisines they serve and the relationship of the ratings with respect to delivery times etc.

Data Overview

Data overview

- Top 5 rows from the dataset

	order_id	customer_id	restaurant_name	cuisine_type	cost_of_the_order	day_of_the_week	rating	food_preparation_time	delivery_time
0	1477147	337525	Hangawi	Korean	30.75	Weekend	Not given	25	20
1	1477685	358141	Blue Ribbon Sushi Izakaya	Japanese	12.08	Weekend	Not given	25	23
2	1477070	66393	Cafe Habana	Mexican	12.23	Weekday	5	23	28
3	1477334	106968	Blue Ribbon Fried Chicken	American	29.20	Weekend	3	25	15
4	1478249	76942	Dirty Bird to Go	American	11.59	Weekday	4	25	24

Data Overview

- Answer 1 : There are **1898 rows** , **9 columns** in the Dataset
- Answer 2 : The datatypes of the different columns in the dataset are:

```
Data columns (total 9 columns):
 #   Column                 Non-Null Count  Dtype  
---  ------                 --------------  -----  
 0   order_id               1898 non-null   int64  
 1   customer_id            1898 non-null   int64  
 2   restaurant_name        1898 non-null   object 
 3   cuisine_type           1898 non-null   object 
 4   cost_of_the_order      1898 non-null   float64
 5   day_of_the_week        1898 non-null   object 
 6   rating                 1898 non-null   object 
 7   food_preparation_time  1898 non-null   int64  
 8   delivery_time          1898 non-null   int64  
dtypes: float64(1), int64(4), object(4)
memory usage: 133.6+ KB
```

Data Overview

Data Types & Column Descriptions

Columns Name Remarks	Data Type	Column Description
order_id	int64	Unique ID of order
customer_id	int64	Unique customer ID will be useful to find not frequent customer
restaurant_name	object	Name of the restaurant will be useful to see popular restaurants, ratings of the restaurants
cuisine_type	object	Cuisine type of useful to see popular cuisine, ratings, cost
cost_of_the_order	float64	cost of the order is the main driver for the business revenue
day_of_the_week	object	Day of the week to find the trend, behavior, when customers may be ordering more etc.
rating	object	Rating indicate customer experience
food_preparation_time	int64	Time it takes for the restaurant to prepare the food on receipt of the order and handover to delivery
delivery_time	int64	Time it takes to deliver food to customer after received from restaurant

Data Overview

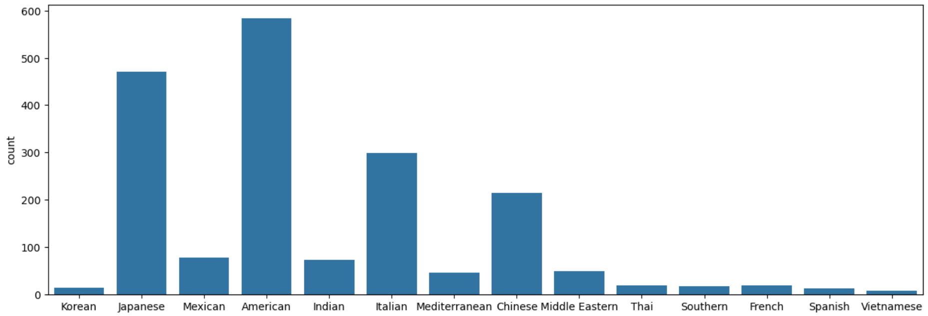
- Answer 3 : There is **no missing data** in any of the columns, so there is no need to fill any missing values

- Answer 4: Food Preparation takes minimum **20** minutes average it takes **27.37** minutes and maximum **35** minutes

- Answer 5: **736** orders are not rated.

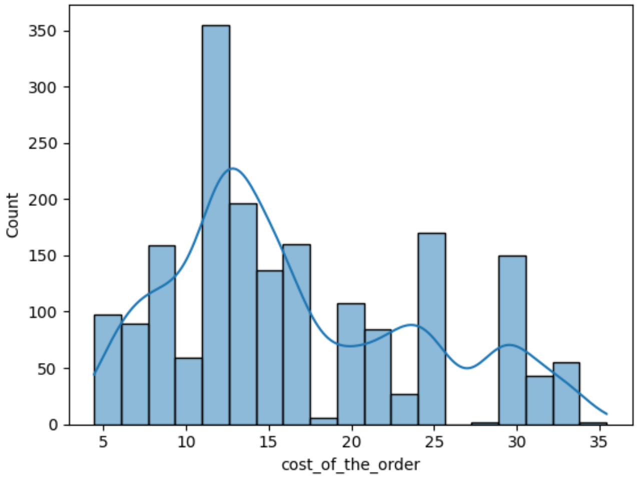
Univariate Analysis

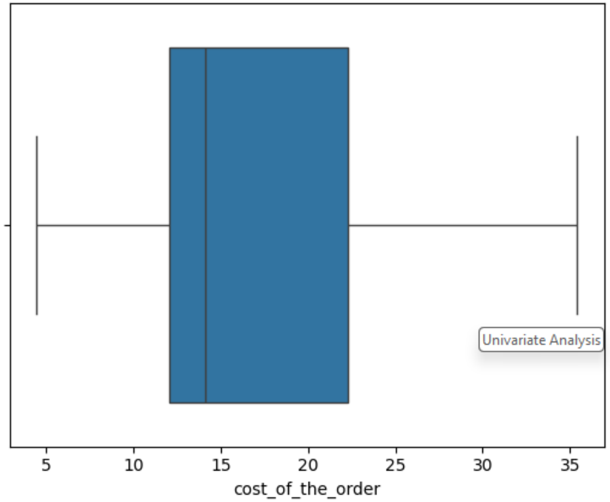
- Answer 6
  - There are 1898 orders by order_id and 1200 customers by customer_id. Since these are ID fields, we will not analyze these.
  - There are 178 restaurants in total serving 14 different types of cuisine
  - Analysis of cuisine_type shows the most popular cuisines are American, Japanese, Italian and Chinese



Univariate Analysis

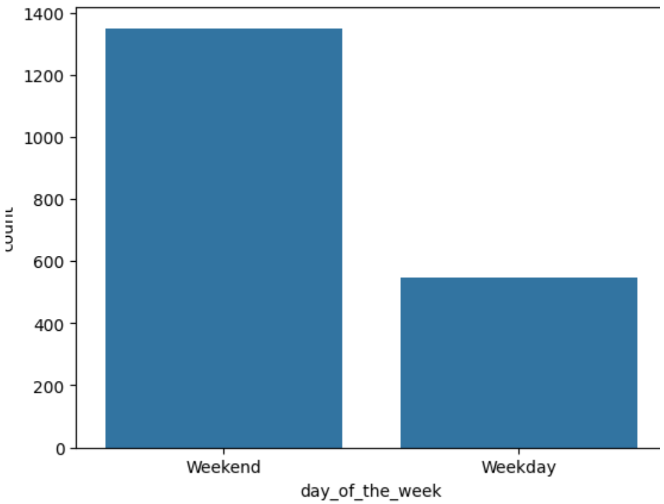
- **Cost of the order** analysis shows that most of the orders are between 12-22.50, with a median value of 14. The data is right skewed with 75% of the orders below 22.50





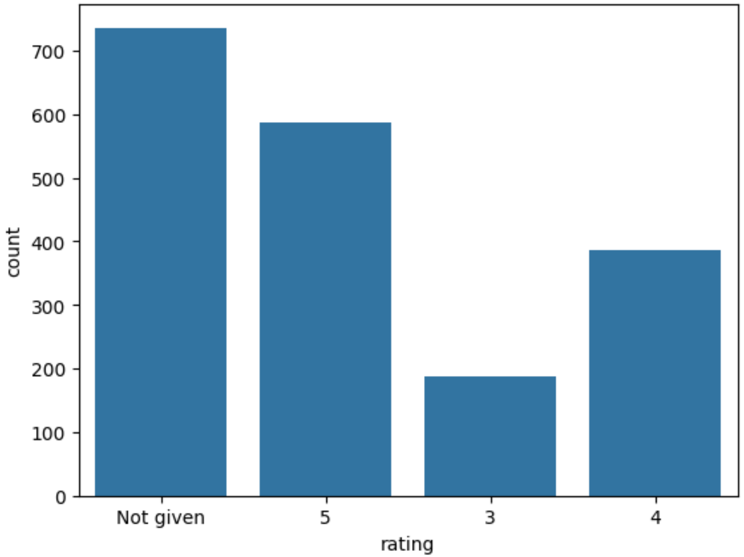
Univariate Analysis

- Day of the week : There are more (2X+) orders being placed over the Weekend than over the weekday



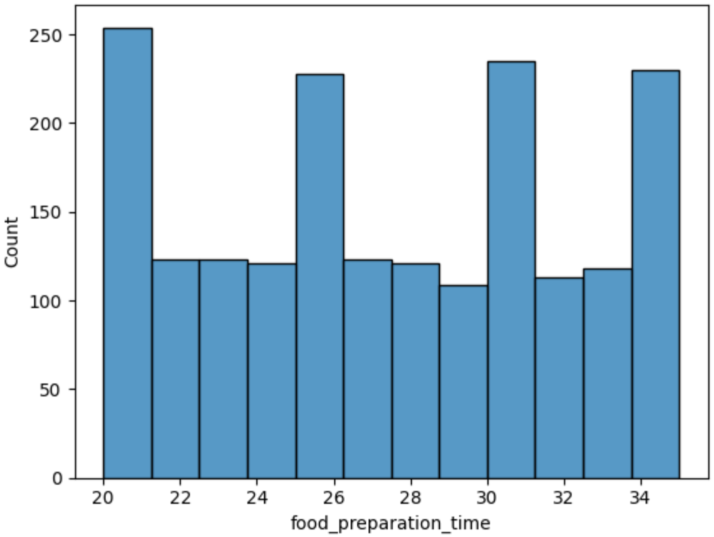
Univariate Analysis

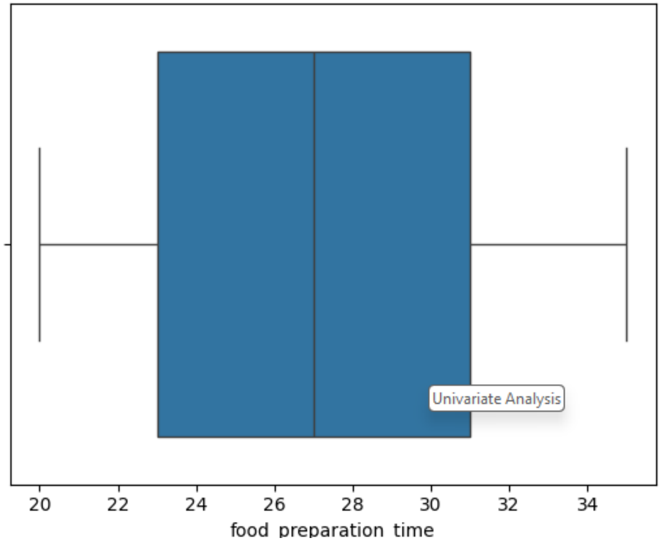
- Rating – A lot of the orders (over 50%) are not rated, the highest rating is 5 and lowest rating is 3.



Univariate Analysis

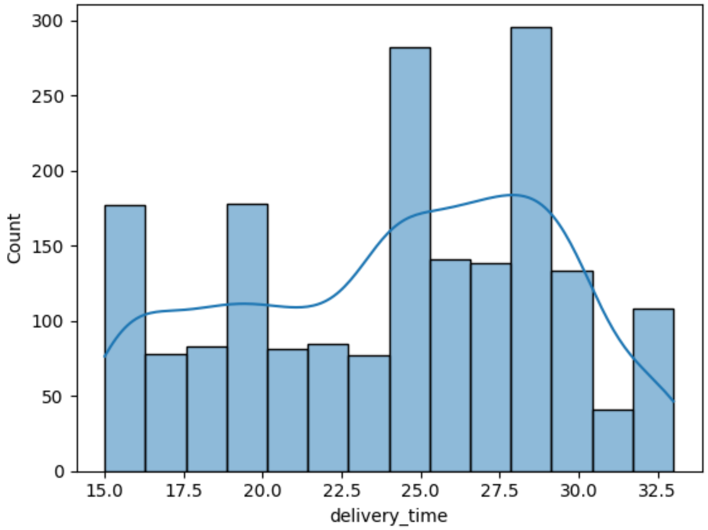
- **Food Preparation time** is evenly skewed and mean food preparation time is 27.37 min

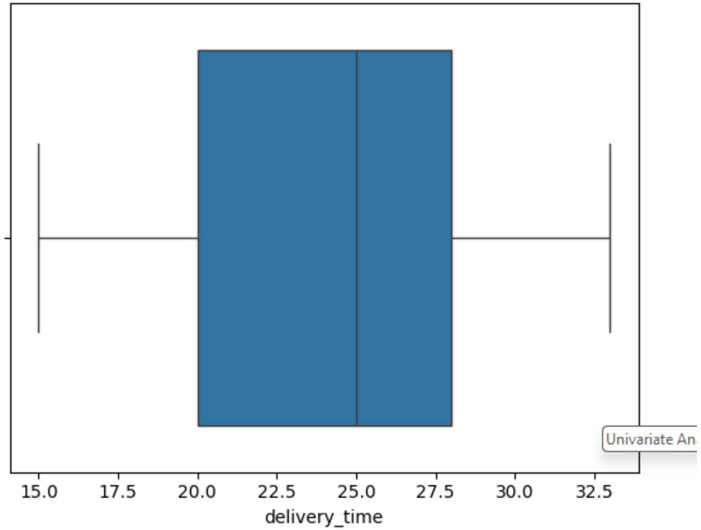




Univariate Analysis

- **Delivery time** has a mean 24.16 mins and 75% orders takes less than 28 min. Data is multimodal with multiple peak as 15, 20, 25 and 30 min, data is left skewed





Univariate Analysis

- Answer 7 : top 5 restaurants in terms of the number of orders received

Shake Shack	219
The Meatball Shop	132
Blue Ribbon Sushi	119
Blue Ribbon Fried Chicken	96
Parm	68

- Answer 8: American Cuisine is the most popular cuisine on the weekends.
- Answer 9: 29.24 % of orders are above 20 dollars.
- Answer 10: The mean delivery time for this dataset is 24.16 minutes.

Univariate Analysis

- Answer 7 – Top 5 restaurants by orders are

restaurant_name	count
Shake Shack	219
The Meatball Shop	132
Blue Ribbon Sushi	119
Blue Ribbon Fried Chicken	96
Parm	68

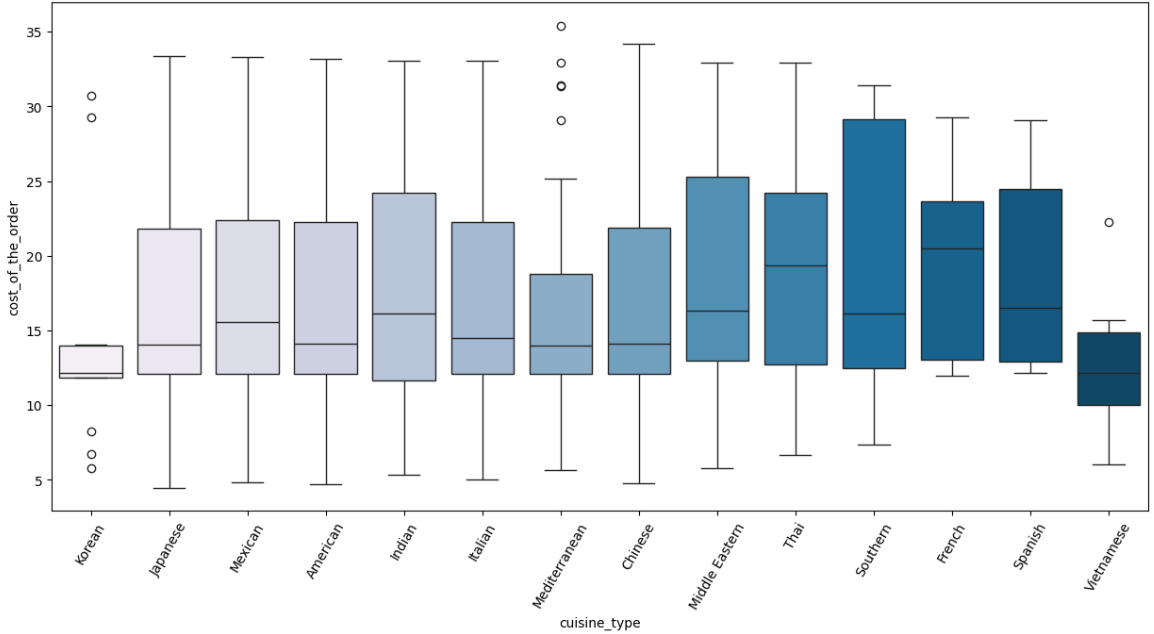
Univariate Analysis

- Answer 8: American Cuisine is the most popular cuisine on the weekends.
- Answer 9: 29.24 % of orders are above 20 dollars.
- Answer 10: The mean delivery time for this dataset is 24.16 minutes.
- Answer 11: Top 5 frequent customers eligible for the 20% discount vouchers

Customer	Order Count
customer_id	
52832	13
47440	10
83287	9
250494	8
259341	7

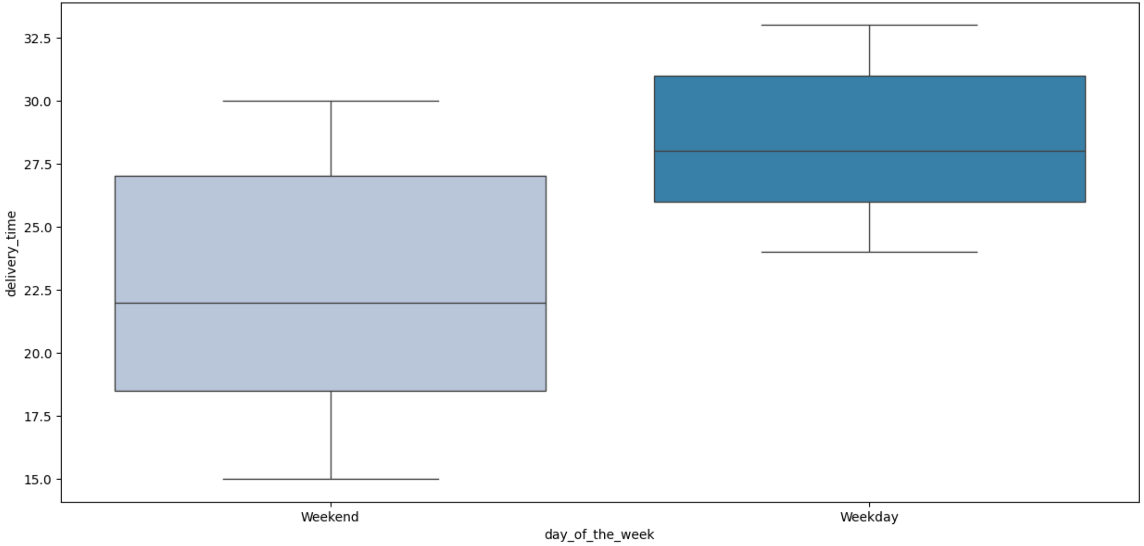
Multivariate Analysis

- Cuisine vs Cost of the order : Southern cuisine has some of the highest cost orders. But the median price is lower than other cuisines. French and Thai cuisine seems to more expensive based on median cost of orders across cuisines.



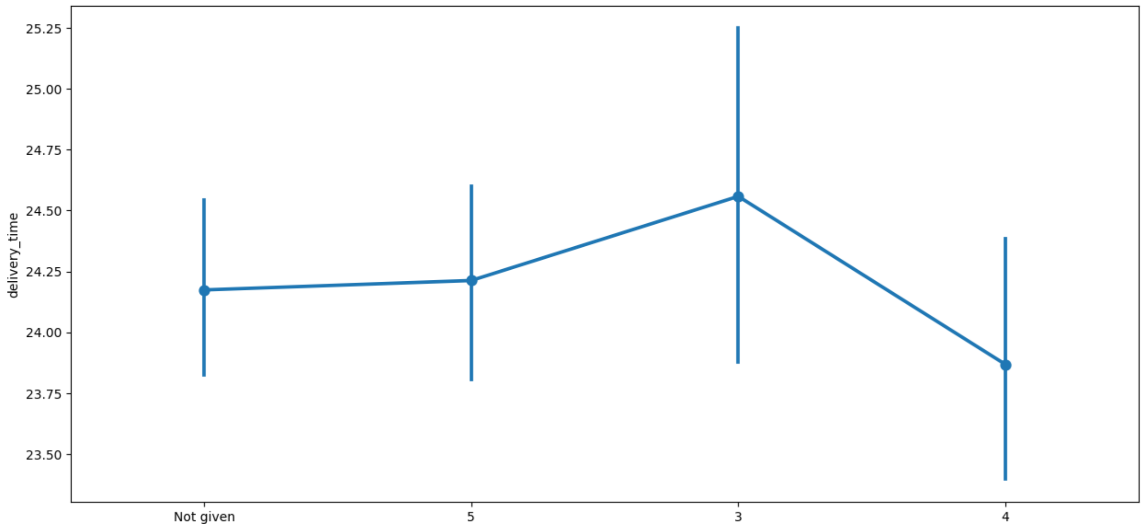
Multivariate Analysis

- Day of the Week vs Delivery time : The delivery times over the weekend are shorter than over the weekday.



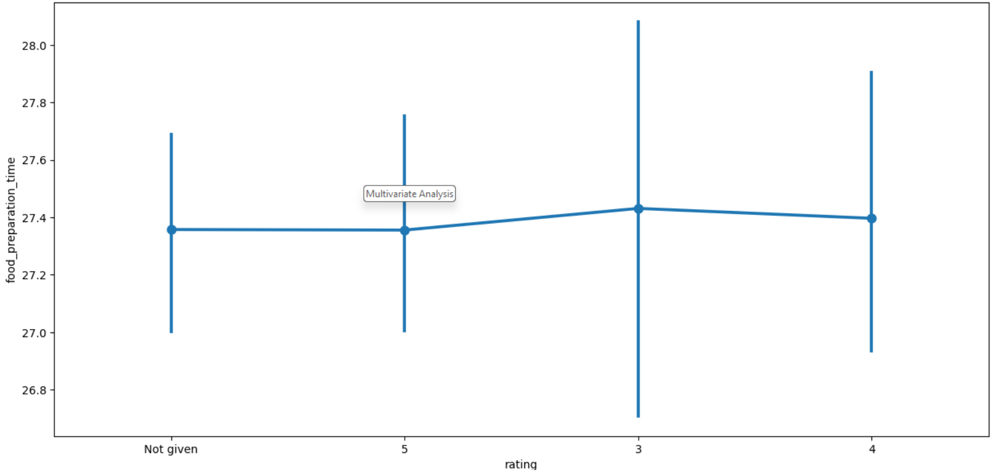
Multivariate Analysis

- Rating vs Delivery time : Ratings seems to drop a bit with increase in delivery times, but it is a **weak correlation** as time for orders with ratings of 5 is a bit higher than the ones rated four.



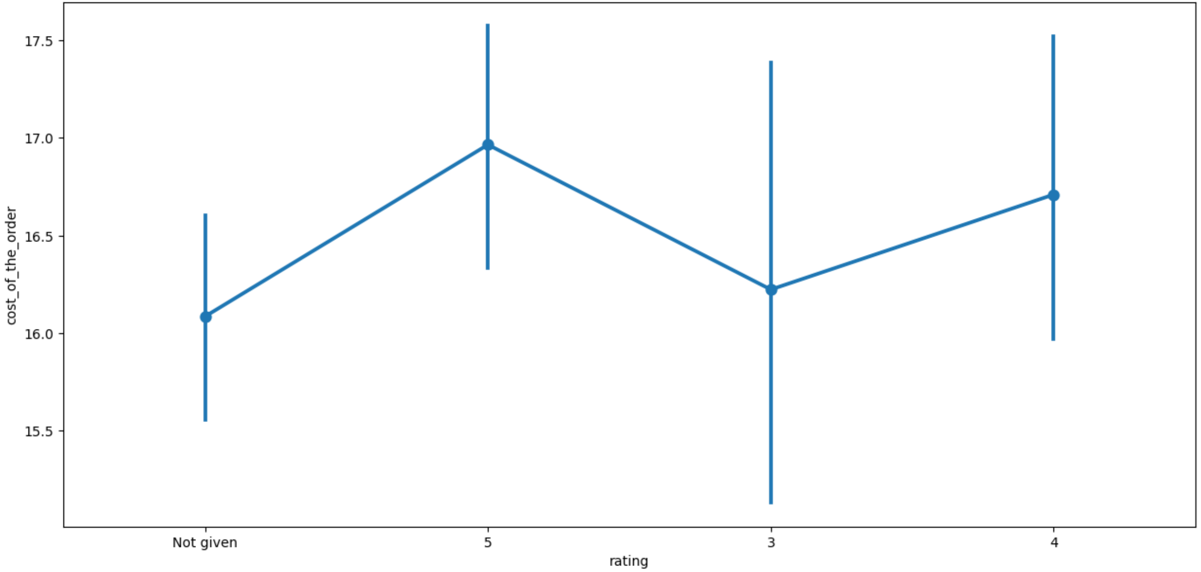
Multivariate Analysis

- Rating vs Food preparation time : Ratings seems to drop a bit with increase in Food preparation times, but it is a **very weak correlation** as ratings are mostly flat with only a slight change to it .



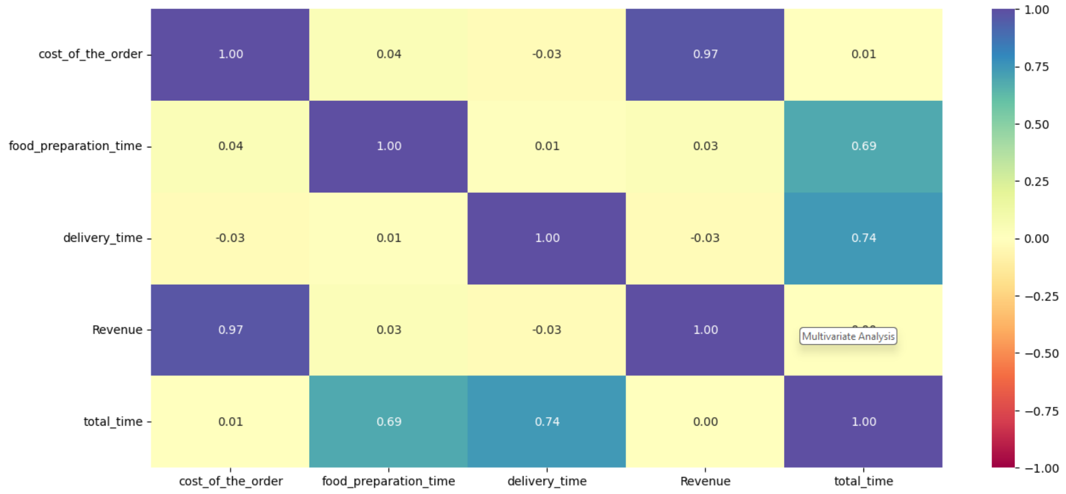
Multivariate Analysis

- Rating vs Cost of the order : There does not seem to be much of a correlation between ratings and cost of the order.



Multivariate Analysis

- Heatmap Analysis



1. total_time is sum of food_preparation and delivery_time and show high correlation

2. Revenue is directly related to the cost of the order, so it also shows high correlation

3. food_preparation and cost_of the order seem to have very weak correlation

4. delivery_time and cost_of the order seem to have very weak correlation

Multivariate Analysis

- Answer 13 : Here are the restaurants having a rating count of more than 50 and the average rating greater than 4 that are eligible for the promotional offer.

	Restaurant Name	Rating
0	The Meatball Shop	4.511905
1	Blue Ribbon Fried Chicken	4.328125
2	Shake Shack	4.278195
3	Blue Ribbon Sushi	4.219178

Multivariate Analysis

- Answer 14 : Net revenue generated by the company across all orders is around 6166.3 dollars
- Answer 15 : 10.54% of the orders cost more than 60 minute. Some more investigation needed to find the reason for high total time it took for these orders. Some additional data points like distance or complexity of the order will be useful.
- Answer 16 : The mean delivery time on weekdays is around 28 minutes. The mean delivery time on weekends is around 22 minutes.

Multivariate Analysis

- **Question 17:** My conclusions from the analysis are:

The analysis of Foodhub data analysis provided key information with respect to the cuisines that are popular among users, and which restaurants are bringing in the most revenue. The analysis also signals that the users order more over the weekend than during the weekend.

- What recommendations would you like to share to help improve the business?

Recommendations :

  - **Cuisine** : Since American was the most popular cuisine, the company could target more revenue by partnering with more American restaurants
  - **Weekdays vs Weekend business**: Weekend account for most of the business, while weekdays are slow hours. Company should focus on promotions to boost weekday sales further
  - **Delivery times** : Delivery times during the weekday are more and there is a negative correlation to revenue. So, company should look at how to reduce delivery times during weekdays by partnering with more restaurants to provide more closer options, especially for the popular cuisines
  - **Ratings:** There are a lot of missing ratings, focused initiative to actively collect feedback and bridge this gap is needed.
  - **Additional Data**: Company should collect additional data with order dates/month to further analyze trends in order volume for seasonal changes, areas where most orders are coming from etc. to more accurately target partnerships with additional restaurants to increase revenue

APPENDIX

Great Learning

Happy Learning !